Aimably

Aimably RDS Instance Vertical Autoscaling

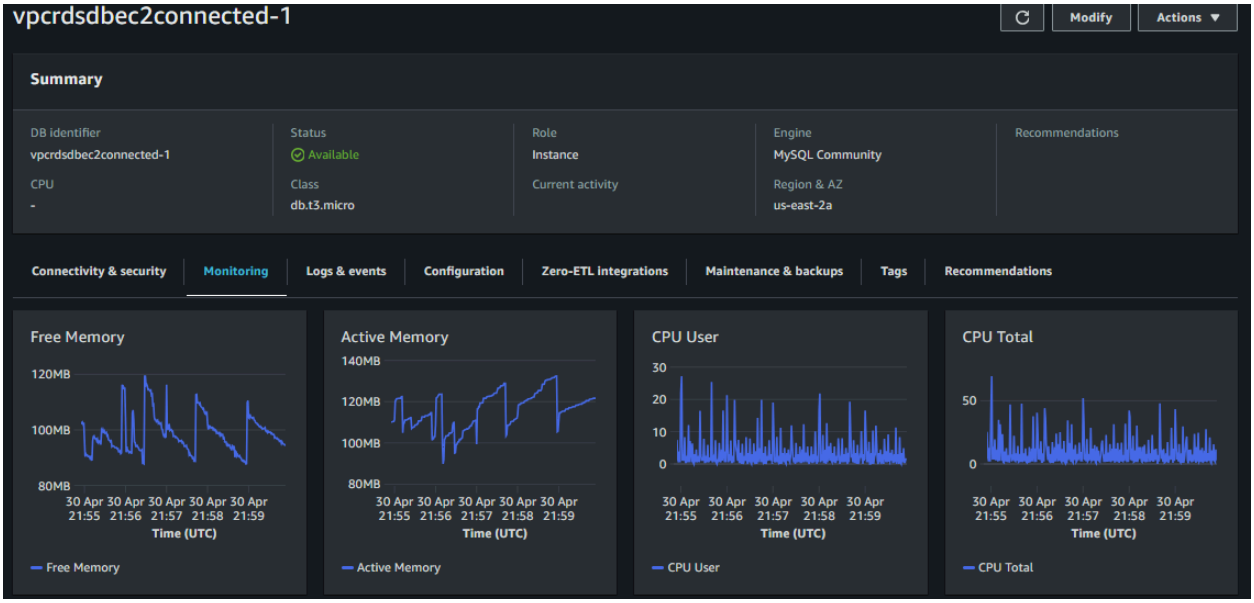
Aimably RDS Instance Vertical Autoscaling is a script-based tool that can automatically increase or decrease the capacity and performance capabilities of RDS instances in response to external factors, such as user demand.

Aimably RDS Instance Vertical Autoscaling includes an AWS Lambda Function that uses AWS Blue/Green Deployment to facilitate vertical autoscaling of RDS instances. The function is triggered based on EventBridge events, which can be triggered by CloudWatch alarms configured to track your chosen performance metrics.

For example, you configure a CloudWatch alarm to alert when the CPU of the RDS instance is below 10% for a specific time period, this can be configured to trigger the Aimably RDS Instance Vertical Autoscaling function to scale down to a lower and less expensive instance class. Conversely, you can configure a CloudWatch alarm to alert if the database CPU is at 90% for a specific period of time, which can trigger the Aimably RDS Instance Vertical Autoscaling function to scale up to a higher and more expensive instance class. This enables you to have the correct instance class based on performance, thereby avoiding expensive overprovisioned databases.

Best Practices

- **Utilization Pattern Review:** Because RDS was not built for vertical autoscaling by default, it is important to become familiar with the utilization patterns of each of your databases to select the best parameters for the Aimably RDS Instance Vertical Autoscaling function.
    - This data is readily available in the Monitoring tab of a database in the RDS console. You can use the standard CloudWatch monitoring or the newer Enhanced Monitoring that is available. You may adjust the reporting time between 1 hour and 15 months to use this data to understand the periods where peak performance is required.



- **Metric Selection:** Every database experiences performance constraints unique to its workload. Select the metrics that become most constrained in order to achieve the best results.

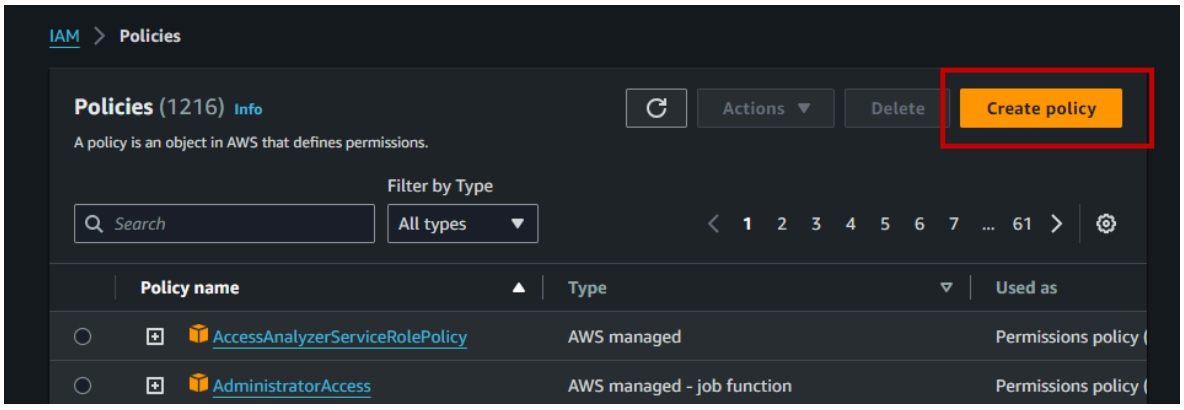
    - Commonly used metrics for an RDS database are CPU Utilization, Database Connections, Freeable Memory, or Read/Write Latency. You can read more in-depth on the various metrics that can be used for an alarm in this support article, AWS CloudWatch/RDS Recommended Alarms.
- **Threshold Selection:** RDS Blue/Green Deployments take an average of 30 minutes or more to complete, which can cause performance issues if thresholds are selected that require immediate results. Select alert thresholds that predict the need for capacity rather than require immediate capacity.
    - In the event of highly predictable traffic patterns, you may also choose to configure events in EventBridge to follow a set schedule instead of using CloudWatch alarms. You can read more in-depth on EventBridge scheduling in this support article, Creating an EventBridge Rule That Runs on a Schedule.
- **Sizing Selection:** The Aimably RDS Instance Vertical Autoscaling function requires that you select two specific instance classes, one to service the lower demand and one to service the higher demand. Make sure to review historical capacity demands and select the instance classes that best represent your needs in both scenarios, ensuring the higher class is not triggered too frequently due to demand on the lower class and that the lower class is not excessively overprovisioned so as to keep costs high.
    - Users may choose to adjust the function to support more than two instance classes.

Prerequisites

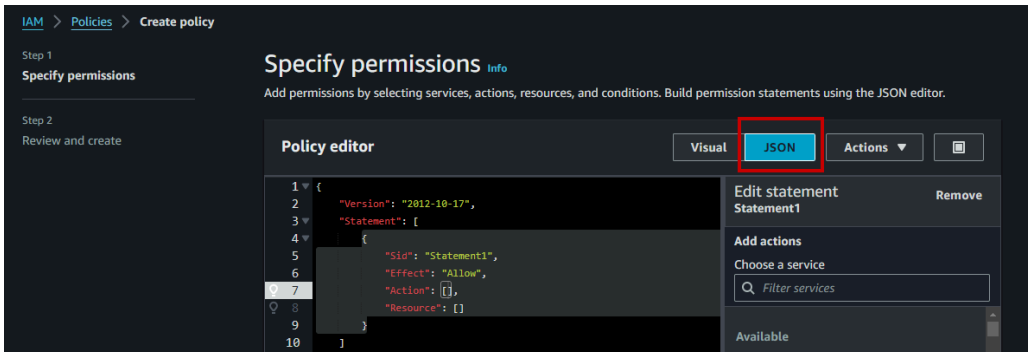
- You need administrator access to your AWS organization
- Your database must make use of Self Managed credential management instead of AWS Secrets Manager
- Aimably RDS Instance Vertical Autoscaling is compatible with Amazon RDS for MySQL, Amazon RDS for PostgreSQL, and Amazon RDS for MariaDB. An alternative solution is provided by Aimably for Amazon RDS Aurora.

Create IAM Autoscaling Policy

- Navigate to the IAM Console in AWS
- Click on the Policies menu item in the left hand side navigation
- Click the Create Policy button in the upper right hand corner



- Click on the JSON button in the Policy Editor title bar



- Paste the below JSON text into the Policy Editor - be sure to copy it from the first bracket to the last

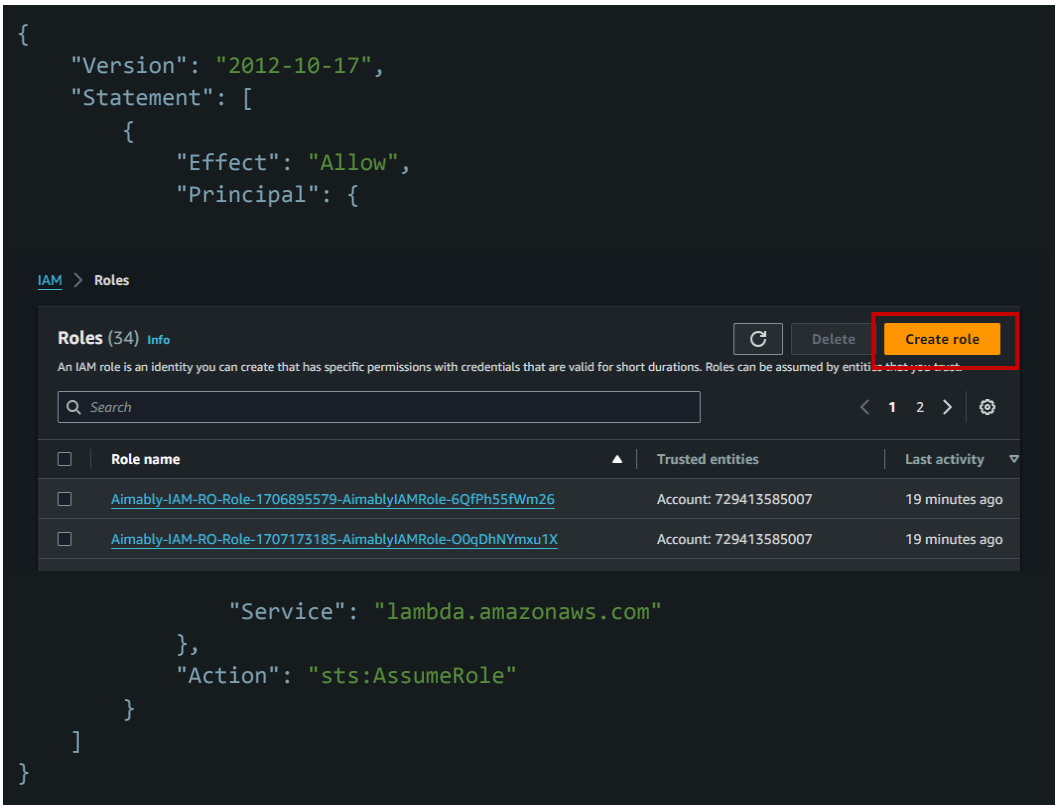
```
{
	"Version": "2012-10-17",
	"Statement": [
		{
			"Sid": "VisualEditor0",
			"Effect": "Allow",
			"Action": [
				"rds:AddTagsToResource",
				"rds:RemoveTagsFromResource",
				"ec2:CreateNetworkInterface",
				"ec2:DescribeNetworkInterfaces",
				"ec2:DeleteNetworkInterface",
				"iam:PassRole",
				"rds:CreateDBInstance",
				"rds:CreateDBInstanceReadReplica",
				"rds:CreateBlueGreenDeployment",
				"rds:DescribeBlueGreenDeployments",
				"rds:SwitchoverBlueGreenDeployment",
				"rds:DeleteBlueGreenDeployment",
				"rds:DeleteDBInstance",
				"rds:DescribeDBInstances",
				"rds:ModifyDBInstance",
				"rds:PromoteReadReplica",
				"rds:DescribeDBClusters",
				"rds:FailoverDBCluster",
				"logs:*",
				"lambda:InvokeFunction"
			],
			"Resource": "*"
		}
	]
}
```

- Click the Next button
- Name the policy `AimablyRDSVerticalAutoscaling`
- Click the Create Policy button

Aimably

Create IAM Autoscaling Role

- Navigate to the IAM Console in AWS
- Click on the Roles menu item in the left hand side navigation
- Click on the Create Role button
- Select the AWS Service radio button
- Under Use Case select Lambda
- Click Next
- Use the search bar and locate and check the following policies:
    - AimablyRDSVerticalAutoscaling
    - AWSLambdaBasicExecutionRole
    - AWSLambdaDynamoDBExecutionRole
- Click Next
- Name the role `AimablyRDSVerticalScalingRole`
- Verify the text in the Trust Policy matches the below JSON text

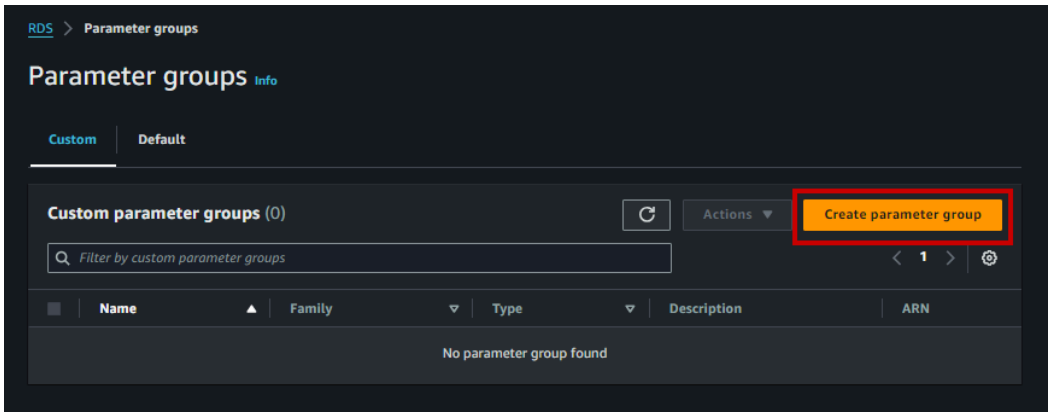


- Click the Create Role button

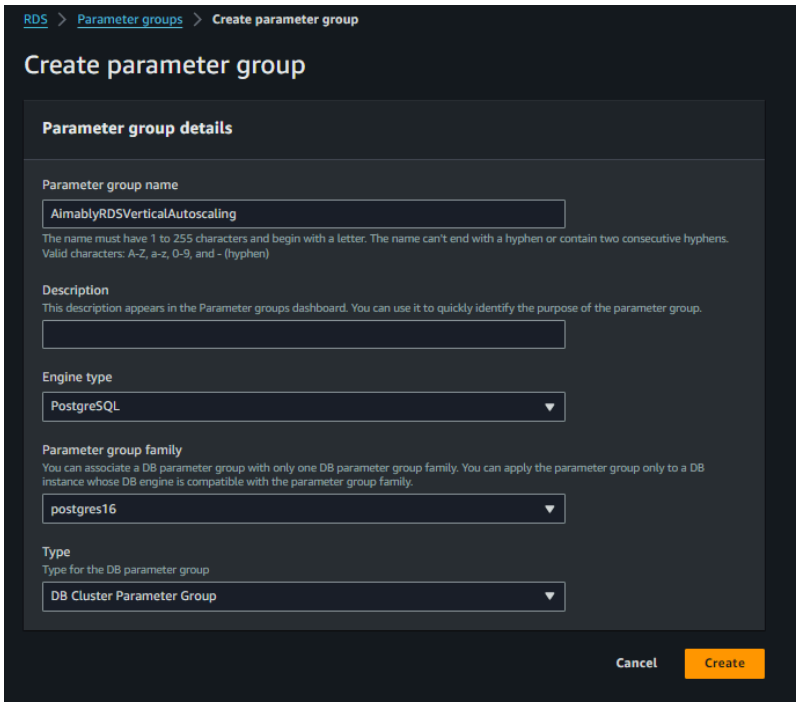
Configure Database Parameter Group

RDS Logical Replication needs to be permitted to make use of Aimably RDS Instance Vertical Autoscaling. By default this option is not enabled for databases and the feature lives in database parameter groups. In order to make use of RDS Logical Replication a custom database parameter group must be created and used.

Aimably



- Navigate to the RDS Console in AWS
- Click on the Parameter Groups menu item in the left hand navigation
- Click Create Parameter Group
- Name the parameter group `AimablyRDSVerticalAutoscaling`
- Input a Description that is meaningful to you
- Select the Engine Type for your database you want to autoscale
- Select the Parameter Group Family (version) for your database you want to autoscale
    - **Please Note**: Select DB Cluster Parameter Group for Type if your database engine is PostGres or MySQL. You may read more about RDS Logical Replication here.

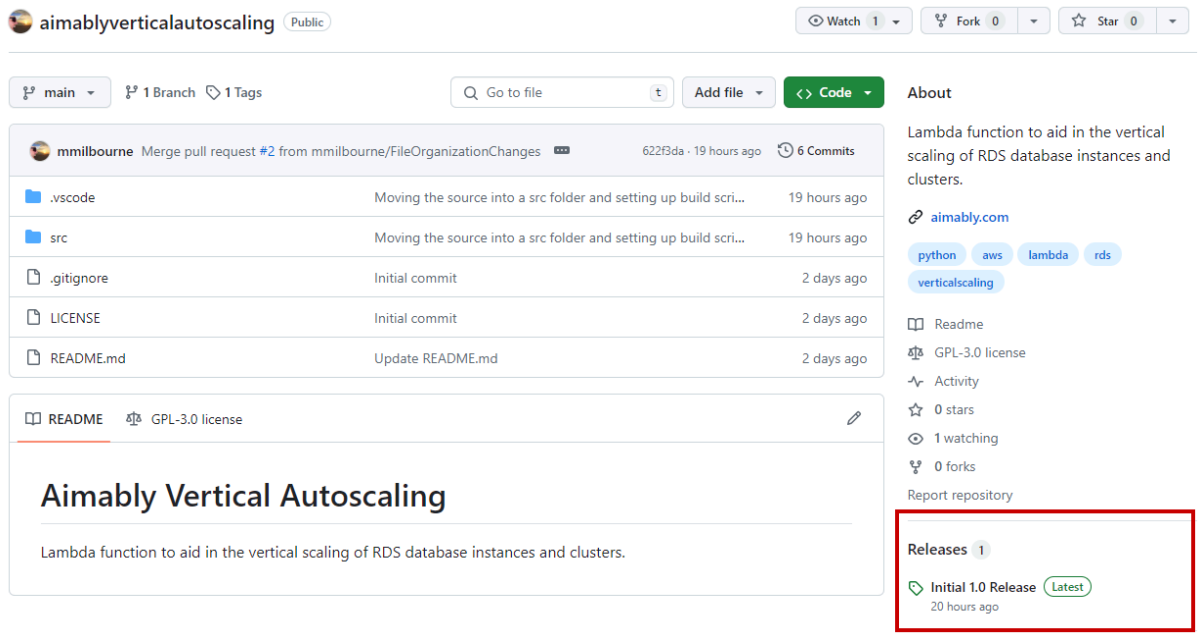


- Click the Create button
- Click on the name of the AimablyRDSVerticalAutoscaling parameter group
- Click the Edit button

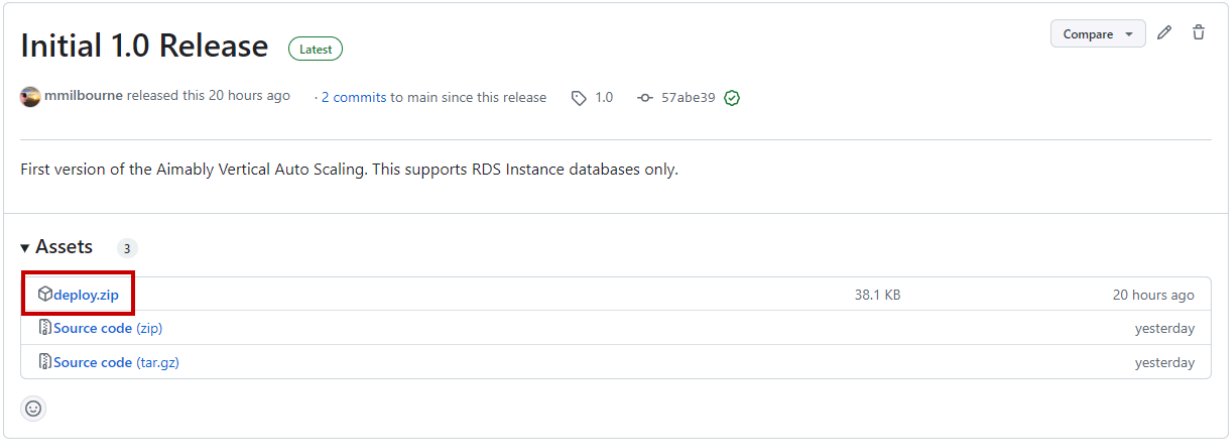
- Enter `rds.logical_replication` into the Filter Parameters text box
- Check the checkbox on the rds.logical_replication parameter
- Click the Save Changes button

Retrieve Source from Github

- Use this link to access the Aimably RDS Instance Vertical Autoscaling repository
- Click on the latest release in the right hand navigation



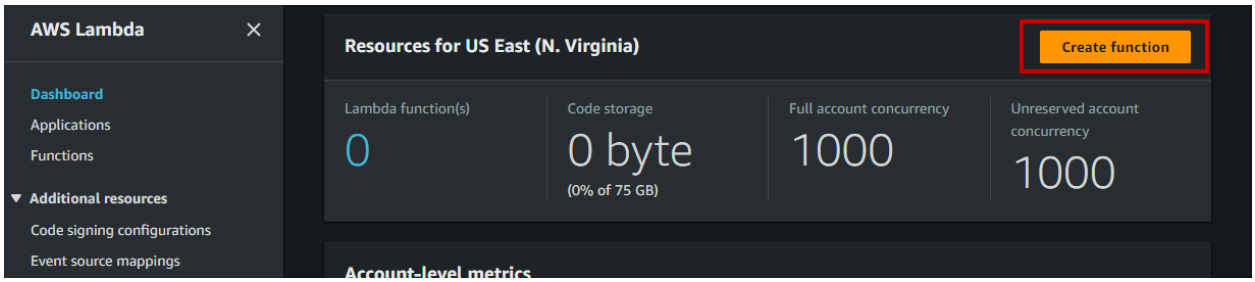
- Click on the file named deploy.zip to download the latest version in a .zip file



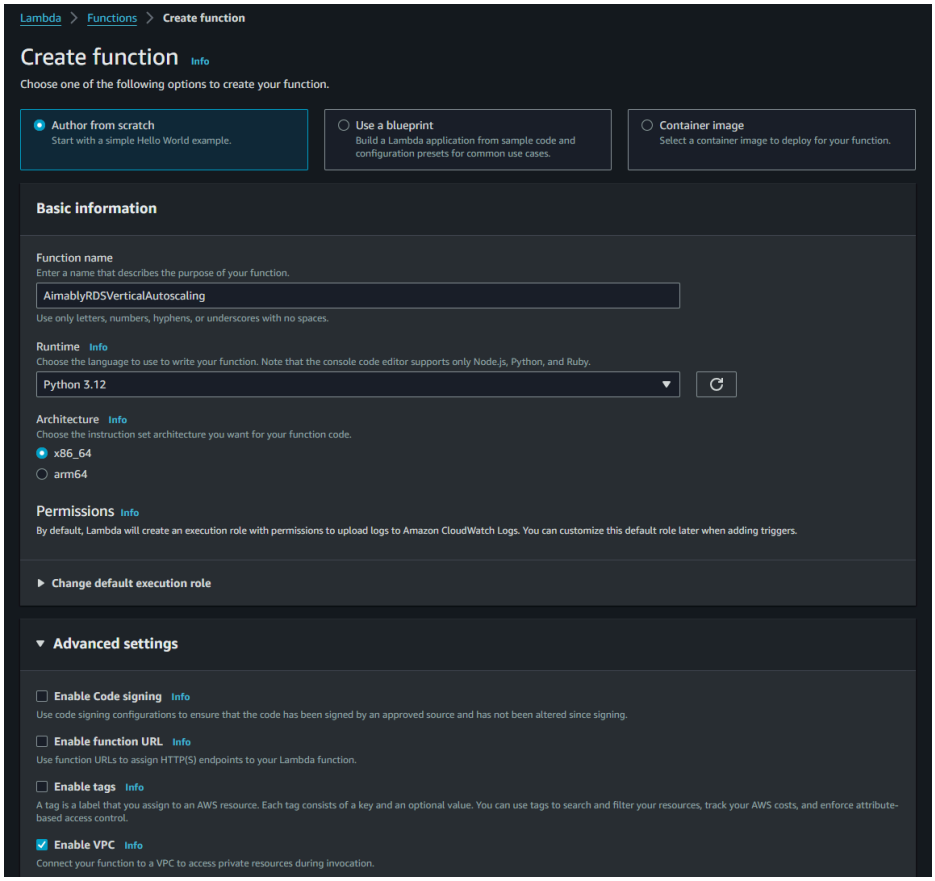
Aimably

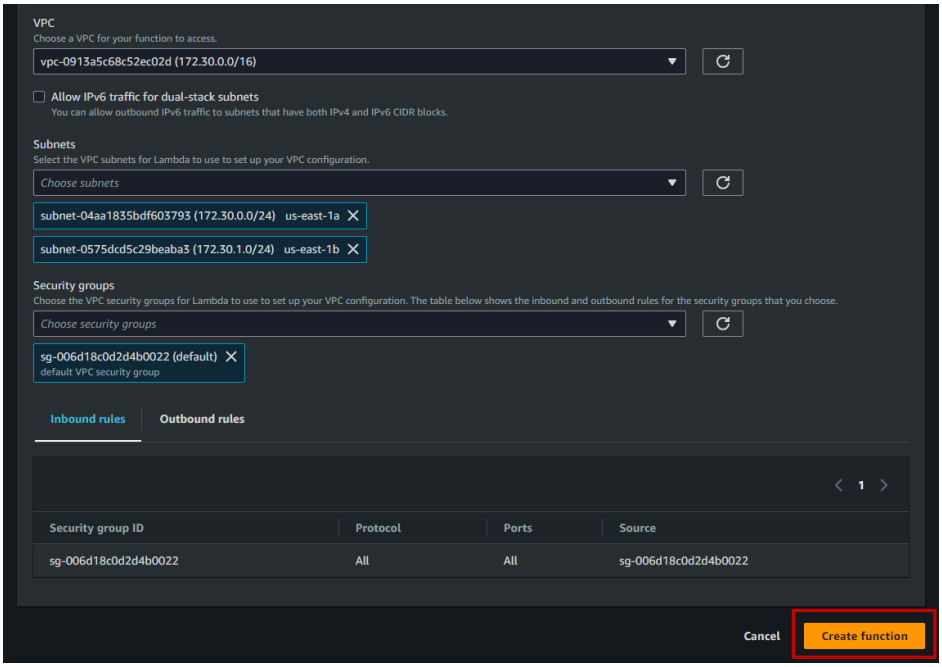
Create Lambda Function

- Navigate to the Lambda Console in AWS
- Click the Create Function button



- Select the Author From Scratch radio button
- Name the function `AimablyRDSVerticalAutoscaling`
- Select the latest version of Python for Runtime
- Select x86_64 for Architecture
- Expand the Change Default Execution Role accordion
- Select Use an Existing Role radio button
- Select the AimablyRDSVerticalScalingRole from the Existing Role drop down
- Expand the Advanced Settings accordion
- Check the checkbox for Enable VPC
- Select the VPC the RDS instance uses in the VPC drop down
- Select the subnet(s) the RDS instance uses in the Subnets drop down
- Select the Security Group(s) the RDS instance uses in the Security Groups drop down





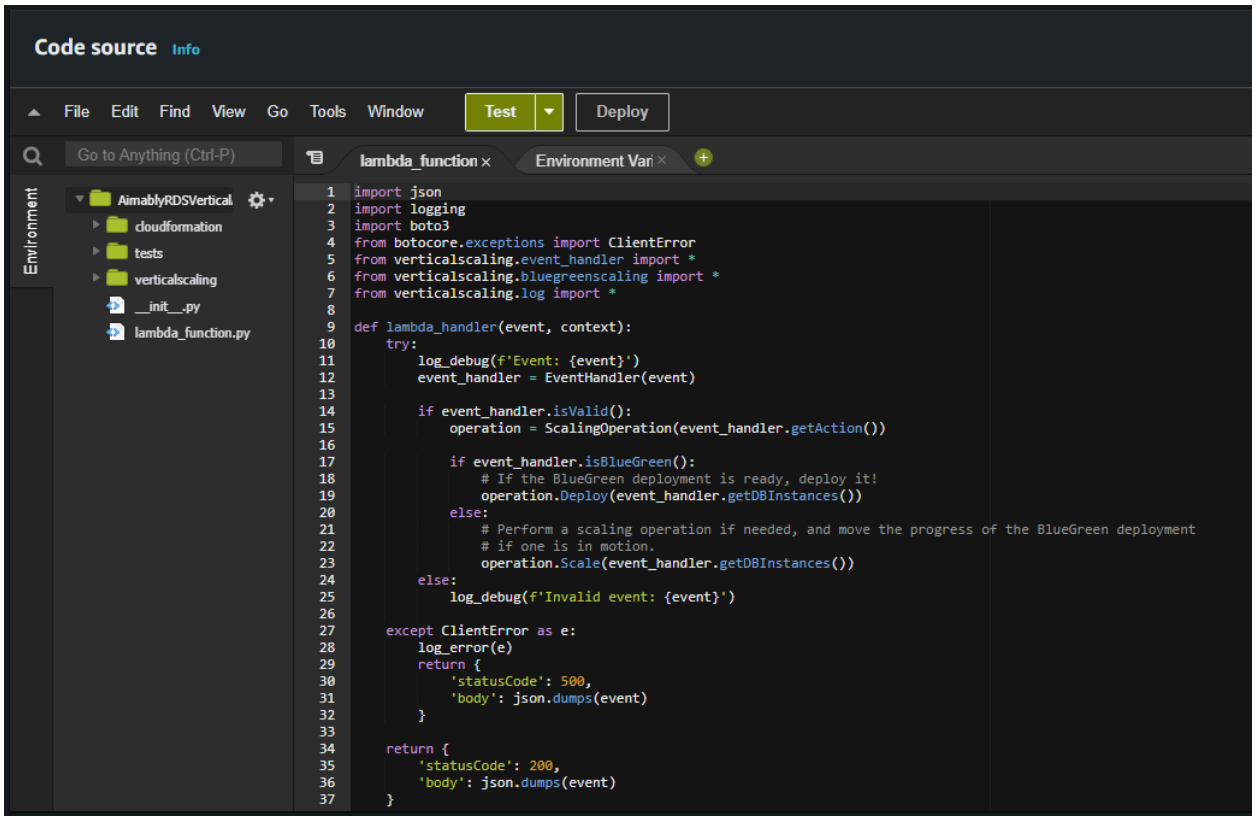
- Click the Create Function button

Create Lambda Layer

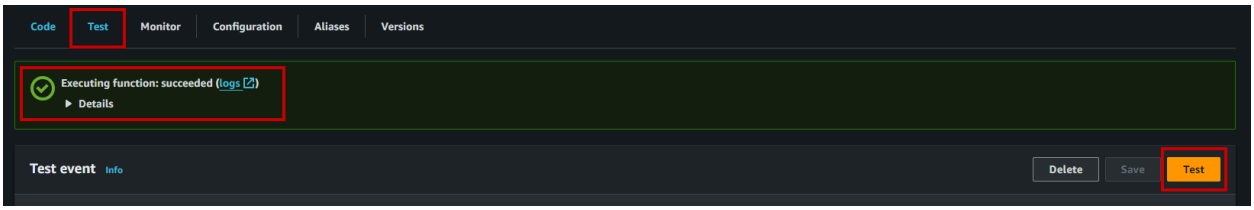
- Scroll to the bottom of the Lambda Function page
- Click on the Add a Layer button
- Select the AWS Layers radio button
- Select the layer named AWSLambdaPowertoolsPythonV2
- Select the latest version in the Version drop down
- Click the Add button

Upload the Source Code

- In the Code Source editor click on the Upload From button and choose .zip file
- Locate the deploy.zip file you downloaded from Github previously
- Drag the deploy.zip file into the Upload a .zip File dialog or click the Upload button and navigate to the file
- Click the Save button
- Verify the code in the Code Source editor matches the below



- Click on the Test tab above the Code Source editor
- Click the Test button



- Verify the Executing function states succeeded and in the Details “Received an unknown event”

Create EventBridge Rules

You will be creating two EventBridge rules to work with Aimably RDS Instance Vertical Autoscaling. These rules will enable the Lambda function to detect the Blue/Green deployment and the necessary state changes to complete the autoscaling events. Please note that EventBridge Rules are region specific. As such you will need to repeat the below steps and create both rules in any region you intend to autoscale in.

- RDSBlueGreen Rule
  - Navigate to the EventBridge Console in AWS
  - Click on the Rules item in the left hand navigation
  - Click the Create Rule button
  - Name the rule `RDSBlueGreen`

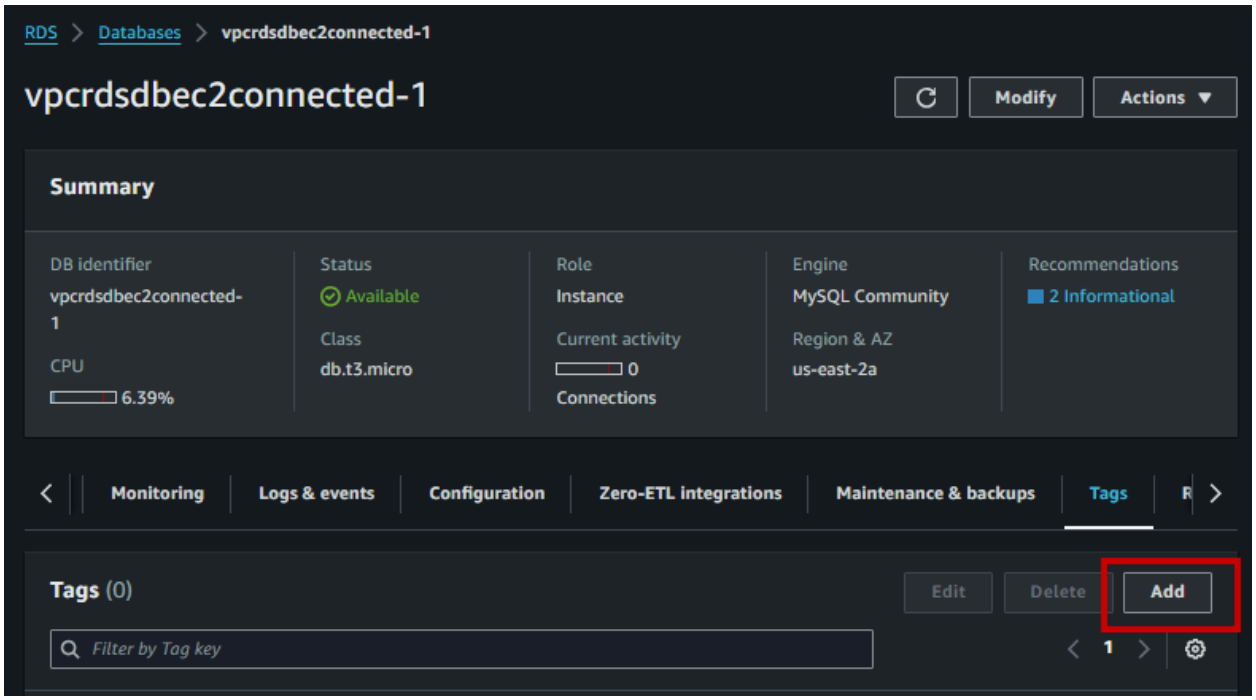
    - Select the default option in the Event Bus drop down
    - Select to enable the rule on the selected event bus
    - Select the Rule With an Event Pattern radio button
    - Click Next
    - Leave all settings on the page as is and navigate to the Event Pattern section at the bottom of the page
    - Select AWS Services in the Event Source drop down
    - Select Relational Database Service (RDS) in the AWS Service drop down
    - Select RDS Blue/Green Deployment Event in the Event Type drop down
    - Click Next
    - Select the AWS service radio button for the Target Type
    - Select Lambda Function from the Select a Target drop down
    - Select the AimablyRDSVerticalAutoscaling Lambda function from the Function drop down
    - Leave all other inputs as they are default configured
    - Click Next
    - Add any tags
    - Click Next
    - Click the Create Rule button
- RDSInstanceStateChange Rule
    - Navigate to the EventBridge Console in AWS
    - Click on the Rules item in the left hand navigation
    - Click the Create Rule button
    - Name the rule `RDSInstanceStateChange`
    - Select the default option in the Event Bus drop down
    - Select to enable the rule on the selected event bus
    - Select the Rule With an Event Pattern radio button
    - Click Next
    - Leave all settings on the page as is and navigate to the Event Pattern section at the bottom of the page
    - Select AWS Services in the Event Source drop down
    - Select Relational Database Service (RDS) in the AWS Service drop down
    - Select RDS DB Instance Event in the Event Type drop down
    - Click Next
    - Select the AWS service radio button for the Target Type
    - Select Lambda Function from the Select a Target drop down
    - Select the AimablyRDSVerticalAutoscaling Lambda function from the Function drop down
    - Leave all other inputs as they are default configured
    - Click Next
    - Add any tags
    - Click Next
    - Click the Create Rule button

Aimably

Configure Database(s)

You need to create two tags on any database you want to use Aimably RDS Instance Vertical Autoscaling with. These tags are used by the Lambda function to understand scale up and scale down events.

- Navigate to the RDS Console in AWS
- Click on the Databases item in the left hand navigation
- Click on the database to access the configuration
- Click on the Tabs tab
- Click the Add button



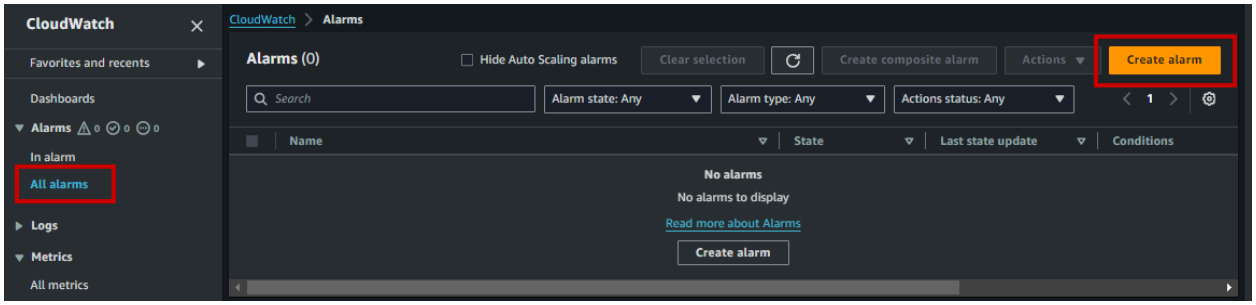
- Tag 1 - Scale Up
  - For the key enter `rds_scaling_high_instanceclass`
  - For the value enter the instance class you want the database to scale up to when it meets conditions, i.e. db.m6gd.xlarge
  - Click the Add button
- Tag 2 - Scale Down
  - For the key enter `rds_scaling_low_instanceclass`
  - For the value enter the instance class you want the database to scale down to when it meets conditions, i.e. db.m6gd.large
  - Click the Add button

Aimably

Create CloudWatch Alarms

As detailed in the Initial Considerations section of this document it needs to be determined what metric you want to use to scale against. Now that you have the metric selected you need to create a CloudWatch alarm to trigger scaling events.

- Navigate to the CloudWatch Console in AWS
- Click on the All Alarms item in the left hand navigation
- Click the Create Alarm button



- Click the Select Metric button
- Click the RDS tile or search for RDS
- Click the DBInstanceIdentifier tile or search for DBInstanceIdentifier
- Search for the metric you want to use for scaling events and check the checkbox next to it
    - Please note you will see the same metric appear multiple times as you must choose it for the RDS instance you want to autoscale
    - For a detailed description of available metrics, please see: https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/rds-metrics.html
- Click the Select Metric button
- Select the desired statistic (Average, Minimum, Maximum etc)
- Select the desired period (10 sec, 30 sec, 1min etc)
- Select your desired threshold type
    - Static - Alarm triggers when the selected metric meets an exact value (i.e. 10)
    - Anomaly Detection - Alarm triggers when the selected metric meets a value in a range (i.e. 1 - 20)
- Select the Alarm Condition (Greater Than, Less Than, Equal etc)
- Enter the Threshold Value
- Click Next
- Click the Add Lambda Action button
- Select the In Alarm radio button for the Alarm State Trigger
- Select the Select Lambda Function radio button
- Select the AimablyRDSVerticalAutoscaling Lambda function
- Click Next
- Name the alarm and include if it is a scale up alarm or scale down alarm
- Click the Create Alarm button

Aimably

Add CloudWatch Alarm Tags

You need to create a tag on each scaling alarm you create so that the Lambda function may read them and use them in concert with the database tags to either scale the database up or scale the database down. So for each metric you create you will add a corresponding tag to represent the appropriate action.

- Navigate to the CloudWatch Console in AWS
- Click on the All Alarms item in the left hand navigation
- Select the alarm created in the previous section's steps by clicking on the name
- Click on the Tags tab
- Click Manage Tags
- Click Add New Tag
- If the alarm you are working on is a scale up alarm then you need to add a scale up tag
    - Enter `rds_scaling_action` as the key (always lowercase)
    - Enter `up` as the value (always lowercase)
- If the alarm you are working on is a scale down alarm then you need to add a scale down tag
    - Enter `rds_scaling_action` as the key (always lowercase)
    - Enter `down` as the value (always lowercase)

Database Scaling Testing

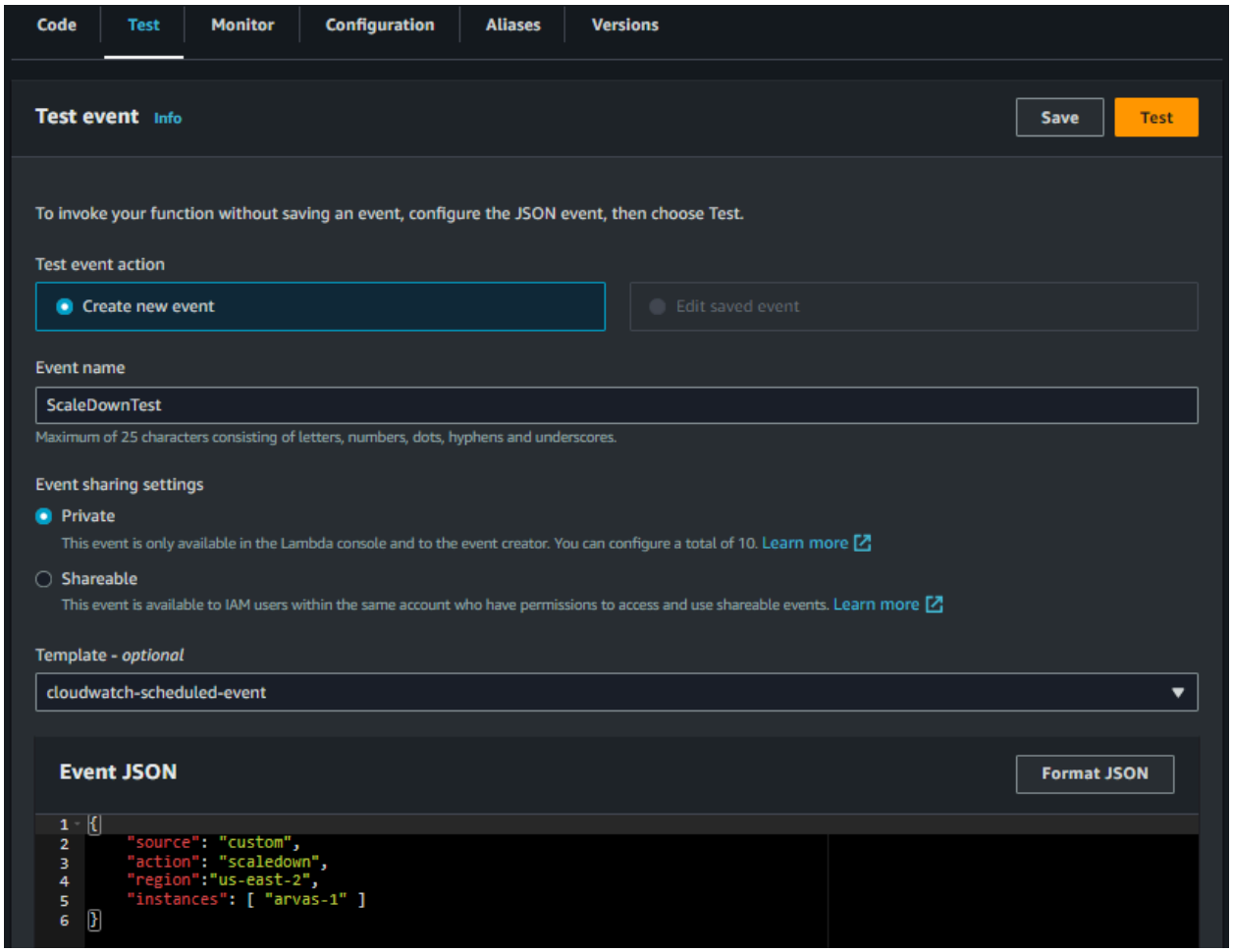
Aimably RDS Instance Vertical Autoscaling is designed to allow you to test scaling events by using custom events to trigger scale up or scale down actions. To test scaling events:

- Navigate to the Lambda Console in AWS
- Click on the Functions item in the left hand navigation
- Locate the AimablyRDSVerticalAutoscaling function and click on the name
- Click on the Test tab above the Code Source editor
- Select Create New Event for the Test Event Action
- Enter an Event Name, i.e. ScaleUpTest or ScaleDownTest
- Select the Private radio button under Event Sharing Settings
- Select all on the existing JSON in the Event JSON editor and delete it
- Copy and paste the below JSON into the Event JSON editor

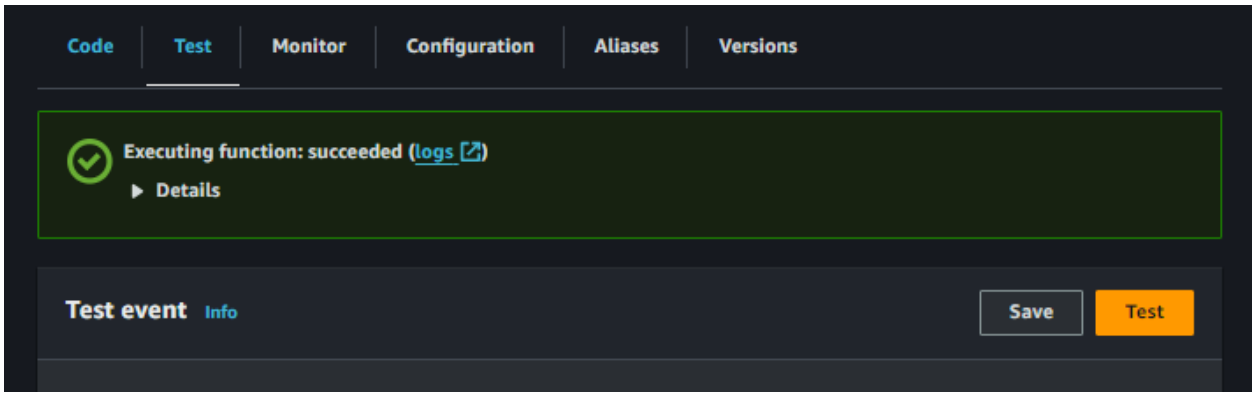
```
{
    "source": "custom",
    "action": "scaleup",
    "region":"us-east-1",
    "instances": [ "ENTER DATABASE INSTANCE NAME" ]
}
```

- You will want to edit the above JSON to define the scaling action you want to test
    - Enter scaleup for a scale up test or scaledown for a scale down test
    - Enter the region of the RDS instance you want to test
    - Enter the database instance identifier of the RDS instance you want to test
    - Click Test

Aimably



- Check for the Executing function: succeeded message



- Verify the database instance class is updated in the RDS Console

- You may also test using the AWS Command Line Interface (AWSCLI) using the following command
    - Edit the command text to reflect the correct:
        - Action - either `scaleup` or `scaledown` depending on which you are testing
        - Region - enter the correct region for the RDS instance database you are testing, i.e. `us-east1`, `us-west-2` etc
        - Instances - enter the DB instance identifier for the database you are testing

```
aws lambda invoke --cli-binary-format raw-in-base64-out --function-name
AimablyRDSVerticalAutoscaling --invocation-type Event --payload '{"source": "custom",
"action": "ENTER scaleup or scaledown", "region":"ENTER REGION", "instances": [ "ENTER
DATABASE INSTANCE NAME" ] }' response.json
```